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Generative AI in Game Development: A Bibliometric Study with an Auditable Computing Workflow

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Abstract

This paper presents a bibliometric mapping of generative AI in game development from 2017 to 2025, covering 1,103 publications retrieved under a dual-axis query of model families (e.g., GANs, diffusion models, LLMs) and development components (e.g., PCG, game assets, NPCs, engines). Using CiteSpace and VOSviewer, we analyze co-occurrence, co-citation, and cluster structures to reveal the field's evolution from algorithm-driven (genetic/evolutionary) to data-driven (deep learning, GAN) and experience-driven (LLM-NPCs, dynamic difficulty, serious games) trajectories. To meet EI/Scopus expectations for computer-application rigor and reproducibility, we document the end-to-end computing workflow (query to de-duplication to text normalization to co-occurrence/cluster construction), fix random seeds, and report lightweight engineering metrics (step-wise runtime and peak memory). We also list software versions and key parameters (time slicing, Pathfinder pruning, LLR labeling; modularity Q and silhouette S) to make figures and tables auditable. The study consolidates trends, pinpoints core venues and authors, and offers a concise, reusable recipe for bibliometric mining in AI-for-games

CCS Concepts

• **Applied computing** → Computers in other domains; Personal computers and PC applications; Computer games; • **Information systems** → Information systems applications; • **Computing methodologies** → Artificial intelligence; Natural language processing.

Keywords

Bibliometrics, scientometrics, procedural content generation, large language models, clustering

ACM Reference Format:

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1 Introduction

In recent years, driven by technological advancements and evolving user demands, the global gaming industry has maintained steady growth, emerging as a cornerstone of the digital entertainment economy [1]. According to The Paper's *Special Report: 2025 Gaming Technology and Market Trends*, the worldwide gaming market reached \$187.7 billion in 2024. Mobile games maintained their dominance with \$103.1 billion (60%) [2], while console games emerged as the growth engine with 5.5% growth (US\$45.9 billion), and the PC market showed steady development (US\$39.9 billion). According to research data from Jianle Shangbo (168 Report), the market size is projected to approach 3,827 billion yuan by 2031, with a CAGR (Compound Annual Growth Rate) of 12.0% over the next six years [3]. Game development is a comprehensive process encompassing planning, design, programming, art style creation, and sound production [4]. It's noteworthy that under the pressure of continuously growing market demands, modern game development faces diverse technical challenges and innovation needs. In 2018, Rockstar Games co-founder Dan Houser made waves by revealing that developers had to work 100 hours a week to meet the release deadline for *Red Dead Redemption 2*. These phenomena highlight the limitations of traditional game development processes: prolonged development cycles, high labor costs, and immense pressure from continuous content production and operational optimization [5]. Particularly as players' expectations for game quality continue to rise, any flaws or content deficiencies in games may lead to user attrition.

Generative Artificial Intelligence (Generative AI, hereinafter referred to as generative AI) is one of the fastest-growing branches in the field of artificial intelligence [6]. As of 2024, China's generative AI sector has shown strong development momentum, with accelerated innovation in large models. To date, over 490 large models have completed registration with the Cyberspace Administration of China. The technological advantages of generative AI are demonstrated through its content generation capabilities, with applications expanding from text creation to diverse fields including image processing, video production, and even 3D model generation [7]. According to the *Generative Artificial Intelligence Application Development Report (2024)*, China's user base for generative AI products reached 230 million by June 2024. The country has established a relatively comprehensive artificial intelligence industry system, with over 4,500 related enterprises across the value chain that covers critical components from chips and algorithms to data infrastructure, platforms, and end-user applications. In the realm of game development, generative AI technology significantly

enhances interactive features within gaming systems while providing efficient solutions for developers' automated asset generation and gameplay mechanism design [8]. By enabling personalized realization of game creativity, this technology demonstrates substantial technical application value and promising development potential.

With generative AI, developers can easily implement the following features: (i) The first innovation is game asset and level auto-generation [9]. In game development, particularly for high-quality productions, creating art assets, world environments, and NPCs remains the most resource-intensive phase, accounting for significant R&D costs. Generative AI is transforming traditional development workflows by automating 2D/3D asset creation, scene assembly and even narrative scaffolding, enabling smaller teams to deliver near-production quality content with fewer resources. (ii) The second innovation is the intelligent NPC behavior and dialogue system [10]. Anuttacon, an AI company founded by Cai HaoYu, co-founder of Mihaly, has launched its new game *Whispers from the Star*. This groundbreaking project features AI-driven character dialogues, with real-time computation for character interactions, emotional expressions, and action sequences, delivering players an unprecedented interactive experience. (iii) The third innovation lies in personalized game content generation. Through AI technology, games can analyze player behavior and habits to create more tailored gaming experiences [11]. In Entertainment Dynamics' upcoming game *MIR 5*, the BOSS Astrion features NVIDIA ACE-powered AI with real-time learning, memory retention, and strategic adjustment capabilities. It analyzes player actions and prioritizes attacking vulnerable targets.

Research on generative AI applications in game development demonstrates distinct interdisciplinary characteristics, spanning computer science, game design, artificial intelligence, and human-computer interaction. Current studies exhibit insufficient practical implementation [12]. While individual case studies and technical applications have demonstrated notable effectiveness, there remains a lack of systematic integration and optimization across the entire development process. This paper aims to systematically summarize the application status of generative AI in game development, analyze its transformative impact on the game development process, discuss the future development direction, and provide theoretical guidance and practical reference for the effective application of generative AI in game development. Through the above work, this paper makes the following contributions: (i) It systematically reveals how emerging findings—particularly the evolution of generative AI from algorithm-driven approaches (genetic and evolutionary computation) to data-driven (deep learning, GANs) and experience-driven paradigms (LLM-based NPCs and adaptive difficulty)—respond to and expand upon classical theories such as *procedural content generation (PCG)*, *experience-driven design*, and *computational creativity*. These findings clarify that contemporary generative AI research not only inherits the algorithmic logic of early procedural generation but also reshapes it toward user experience modeling and co-creative interaction, marking a theoretical shift from rule-based generation to adaptive and player-centered intelligence. (ii) It outlines a technical framework linking generative AI to major stages of game development—asset creation, level design, narrative generation, and NPC systems—providing a

structured view of technological integration. (iii) It offers developmental insights for generative-AI practitioners by highlighting cross-domain opportunities such as multimodal fusion and interactive storytelling. (iv) It proposes transformation pathways for the game industry, emphasizing how generative AI supports intelligent production, efficiency improvement, and sustainable innovation.

The structure of this paper is as follows: Section 2 summarizes the research methods and basic bibliometric analysis; Section 3 discusses the collaborative network and shows the visualized chart of the collaborative network; Section 4 discusses the co-citation network and its corresponding visualized knowledge graph; Finally, Section 5 provides a summary of this study.

2 Research methods and basic analysis

2.1 Literature selection

This study underwent rigorous screening during journal selection and literature retrieval. In the English literature selection process, we focused on high-quality academic publications in the fields of computer science and artificial intelligence. Core sources included specialized journals such as IEEE Transactions on Games, ACM Transactions on Graphics, and Entertainment Computing, as well as top-tier conferences and journals in the AI domain like NeurIPS, ICML, and IEEE Conference on Computational Intelligence and Games, ensuring coverage of interdisciplinary research achievements [13].

This paper conducts keyword screening based on the core technological systems of generative AI and game development. Given the interdisciplinary nature of this research, the keyword system encompasses not only technical terms in generative AI but also key milestones in game development processes, addressing the complexity of technology application in current industry practices. To achieve this, we established a dual-dimensional framework integrating generative modeling technologies (including “generative AI”, “large language models”, “diffusion models”, GAN, Transformer, etc.) with game development components (“game development”, “procedural content generation”, “game assets”, NPCs, “game engines”, etc.).

Building on this foundation, we conducted an enhanced literature screening. The search was limited to journal articles and conference papers, excluding works directly related to “game theory” to prevent thematic drift. We further filtered out studies solely addressing traditional procedural generation methods, ensuring the results closely aligned with the core focus of the intersection between “generative AI” and “game development”.

Using these target keywords and search criteria, we conducted a literature search in the Web of Science database for English-language publications covering the period from January 1, 2017—September 20, 2025. The process ultimately identified 1,103 research papers on generative AI applications in game development. By selecting Citespace 6.4.1 and VOSviewer as bibliometric analysis tools, we established a robust foundation for subsequent systematic bibliometric studies to reveal the research trajectory and emerging trends of generative AI-driven game development.

2.2 Methodology

2.2.1 Co-creation analysis. Co-occurrence analysis is a research method that reveals internal connections by statistically analyzing the frequency and patterns of co-occurrence among words, concepts, or features in texts [14]. In bibliometrics, this approach is commonly used to identify research hotspots, construct conceptual frameworks, and predict disciplinary trends. Results are typically visualized through network diagrams or heat maps, illustrating relationships between keywords, authors, or institutions. For example, high-frequency keyword co-occurrences indicate closely related research themes, while author co-occurrences reflect the intensity of academic collaboration. In the third part of this study, co-occurrence analysis is used to mine high-frequency keyword combinations in the fields of generative AI and game development (such as “programmed content generation” and “machine learning”) to reveal hot spots and cross patterns of technology application.

2.2.2 Co-citation analysis. Citation co-occurrence analysis, an empirical research method in bibliometrics, examines how multiple papers are mutually cited by subsequent works to reveal the knowledge base and core connections within academic disciplines. The core hypothesis posits that higher citation co-occurrence frequency correlates with greater thematic similarity and academic interconnectivity among publications. This analytical approach can be extended to examine dimensions such as authors and journals, utilizing methods like cluster analysis and multidimensional scaling to map knowledge networks, thereby identifying research frontiers and structural patterns within scientific communities. In the fourth part of this study, co-citation analysis was applied to identify high-impact literature in the field of game development generated by AI, and trace the evolution path of technology development through co-citation network tracking (such as the change of core literature cluster caused by the advent of ChatGPT) [15].

2.3 Descriptive statistics

2.3.1 Annual statistics and growth trend of literature. Through a systematic tracking of literature on generative AI and game development from 2017 to 2025, the annual number of publications was statistically analyzed. As shown in the Figure 1, this line chart clearly illustrates the annual distribution trend of literature quantity in related fields from 2017 to 2025. In 2017, the number of publications in this field was only 38. It saw a slight increase to 56 in 2018 and reached 79 in 2019. From 2020 to 2022, there was steady growth with 83, 85, and 91 papers published respectively. In 2023, a significant surge occurred, with publication volume reaching 117 papers—a 28.6% year-on-year increase.

Notably, the official launch of ChatGPT in November 2022 marked a key turning point. This groundbreaking technology not only accelerated the widespread adoption of generative AI but also directly catalyzed an exponential surge in academic research. The number of publications skyrocketed to a historic peak of 255 in 2024, representing an 117.9% increase from 2023. Although data for 2025 is only available through mid-September, it still shows significant growth compared to 2023 levels, indicating that the field has entered a new phase of stable development. These trajectory

changes foreshadow that the application of generative AI technology will spark substantial technological exploration and production revolution in game development.

2.3.2 Distribution of published journals. This study analyzed publication volumes across journals in the generative AI gaming development field, ranking them by citation count (top 15) to generate the following journal distribution bar chart (Figure 2). The analysis revealed a cluster of core journals with significantly higher publication output. As shown, IEEE Transactions on Games, a leading journal in gaming technology, has long focused on interdisciplinary research spanning game design, artificial intelligence, and interactive modeling. With 46 publications (18.3% of total output), it dominates the field as the most influential core journal. The second tier includes the 2024 IEEE Conference on Games (COG 2024) (an international top-tier conference in AI-driven gaming) and IEEE Transactions on Computational Intelligence and AI in Games, accounting for 15.2% and 12.7% respectively. This distribution highlights the balanced emphasis on academic conferences and specialized journals within the discipline. The dominance of IEEE-affiliated journals and conferences indicates that these publications have become the primary knowledge-sharing platform for generative AI gaming development.

Notably, due to the interdisciplinary nature of this field, cross-disciplinary academic journals such as Entertainment Computing (9.5%) and ACM Transactions on Graphics (8.1%) demonstrate significant academic influence. As a top-tier journal in computer graphics, the high publication volume of ACM Transactions on Graphics indicates that generative AI game development technology fundamentally emphasizes innovation in underlying graphics techniques. This aligns with the technical demands of using generative AI to revolutionize visual rendering and asset creation. In addition, the Foundations of Digital Games conference papers (including multiple conferences from 2020 to 2025) accounted for 21.4% of the total, indicating that the international conference has also become an important platform for publishing research results in this field.

The above distribution pattern shows that generative artificial intelligence has formed an academic communication system in the field of game development with IEEE and ACM series publications as the core, academic journals and conference papers as the emphasis, and a strong interdisciplinary color, showing a strong disciplinary ecology.

3 Cooperative network visualization

3.1 Country cooperation

Figure 3 presents the co-occurrence network map of generative artificial intelligence in the field of game development. The United States and China rank as the top two countries globally with 288 and 159 publications respectively, forming the core research forces in this field. Developed countries such as the United Kingdom (88 publications), Germany (62 publications), and Canada (53 publications) have formed a stable second tier, while emerging research nations like Brazil, South Korea, and Japan have also demonstrated significant vitality.

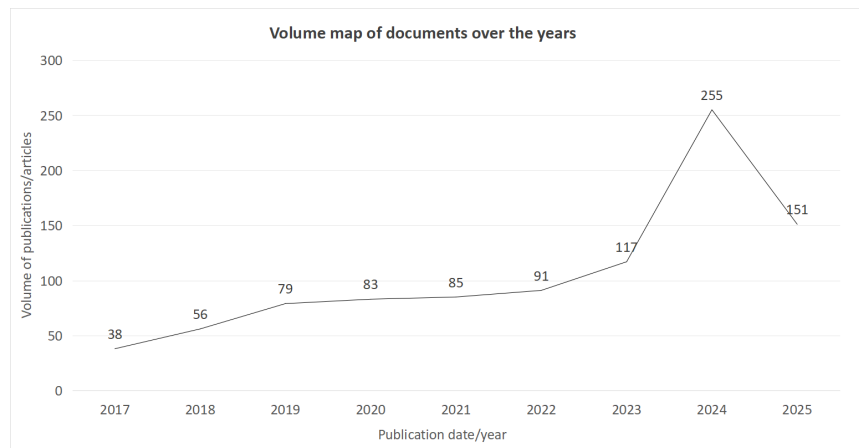


Figure 1: Annual statistical line chart of document volume



Figure 2: Journal distribution bar chart

From a temporal perspective, the color changes of nodes indicate that the United States took the lead in conducting research as early as 2017, followed by China, the United Kingdom, and Germany in 2018, with Canada and Brazil joining in 2019, demonstrating a development trajectory of diffusion from developed countries to emerging nations.

As shown in Table 1, the quantitative analysis of mediation centrality indicators reveals that the United States (0.35) and Canada (0.75) serve as core hubs in the network, while countries like the United Kingdom (0.21) and Germany (0.18) demonstrate significant mediating roles. In Figure 3, national nodes are highlighted with purple circular markers. These highly centralized nations all possess robust economic strength and advanced research infrastructure, illustrating a positive correlation between economic development levels and scientific innovation capabilities.

In addition, the international cooperation network exhibits distinct regional clustering characteristics. The United States maintains cooperative relationships with European countries such as Ireland and Poland, while the United Kingdom collaborates with a broader range of nations including Germany and Brazil. Countries in the Asian region, such as China and South Korea, have also formed regional sub-networks of cooperation. Notably, there is no direct academic collaboration among the United States, the United Kingdom, and China within this network. As research hubs

for their respective communities, they tend to develop their own cooperative networks, forming three significant yet loosely connected academic groups within the network. This phenomenon reflects the regional characteristics present in current international cooperation within this field.

3.2 Institutional cooperation

This study systematically examines the academic influence of generative artificial intelligence (AI) in game development within English-speaking institutions. Based on the narrative summaries derived from bibliometric analysis and the content in Figure 4, the following findings emerge:

In terms of total citation frequency, New York University leads the global rankings with 34 citations, demonstrating the extensive academic influence of its research achievements. The University of London ranks second with 29 citations, followed by the University of Malta in third place with 27 citations. The University of California system and Northeastern University make up the top five with 24 and 22 citations respectively. Notably, two institutions in the top five are American universities, highlighting the significant influence of U.S. academic institutions in this field.

In the centrality ranking of intermediary networks, the University of Ohio system topped the list with a centrality value of 0.07, highlighting its central role in global collaborative networks.

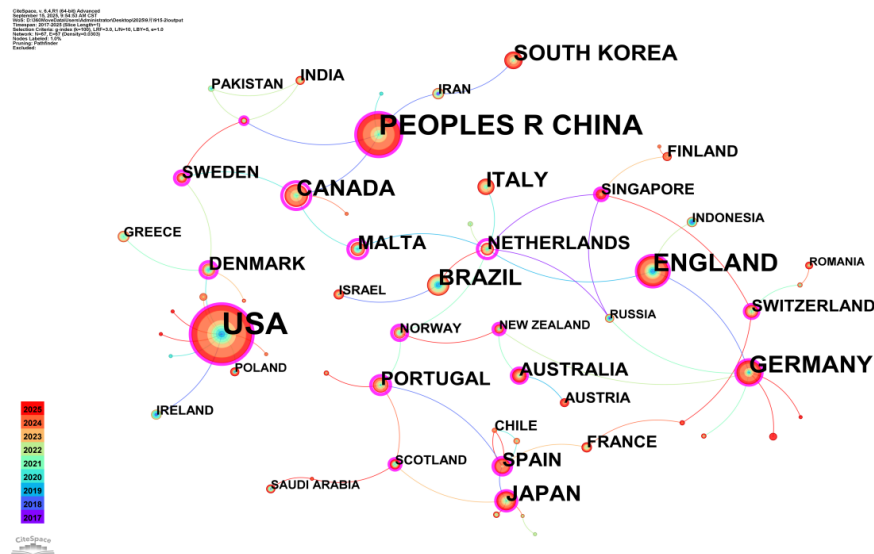


Figure 3: National cooperation networks

Table 1: Profile of the top 15 countries by number of publications

Country	Publications	Collaboration Degree	Centrality
USA	288	8	0.35
PEOPLES R	159	4	0.21
CHINA	88	3	0.1
ENGLAND	62	6	0.18
CANADA	53	4	0.75
BRAZIL	49	2	0.05
SOUTH KOREA	46	1	0
JAPAN	43	4	0.17
ITALY	40	1	0
PORTUGAL	28	4	0.47
MALTA	28	2	0.72
SPAIN	26	5	0.25
DENMARK	25	4	0.46
NETHERLANDS	24	8	0.96

Harvard University, the University of Massachusetts, and Queen Mary University of London tied for second place with a shared centrality value of 0.06. This distribution pattern indicates that knowledge flows exhibit a multi-polar trend within these networks. Notably, the participation level of industry is equally significant. Microsoft Corporation entered the top ten with an betweenness centrality value of 0.05, demonstrating the crucial role of enterprises in industry-academia-research collaborative innovation.

In addition, institutions in Asia also performed well, with the Chinese Academy of Sciences, Tsinghua University, City University of Hong Kong and University of Macau all ranking among the top in various indicators, showing the rapid rise of Asia in this research field.

The forward-thinking distribution pattern demonstrates that generative AI in game development has established a global research network characterized by deep academic-industry integration and collaborative development between Eastern and Western institutions. North American organizations maintain their leading position through traditional strengths, European entities sustain significant influence via close partnerships, while Asian institutions demonstrate remarkable growth potential.

3.3 Author collaboration

As illustrated in Figure 5’s author collaboration network, generative AI has established a prominent academic community in game development. Notably, New York University’s Julian Togelius leads with 29 publications and 17 collaborations, cementing his leadership in the field. Following closely are Antónios Liapis (17 papers) from

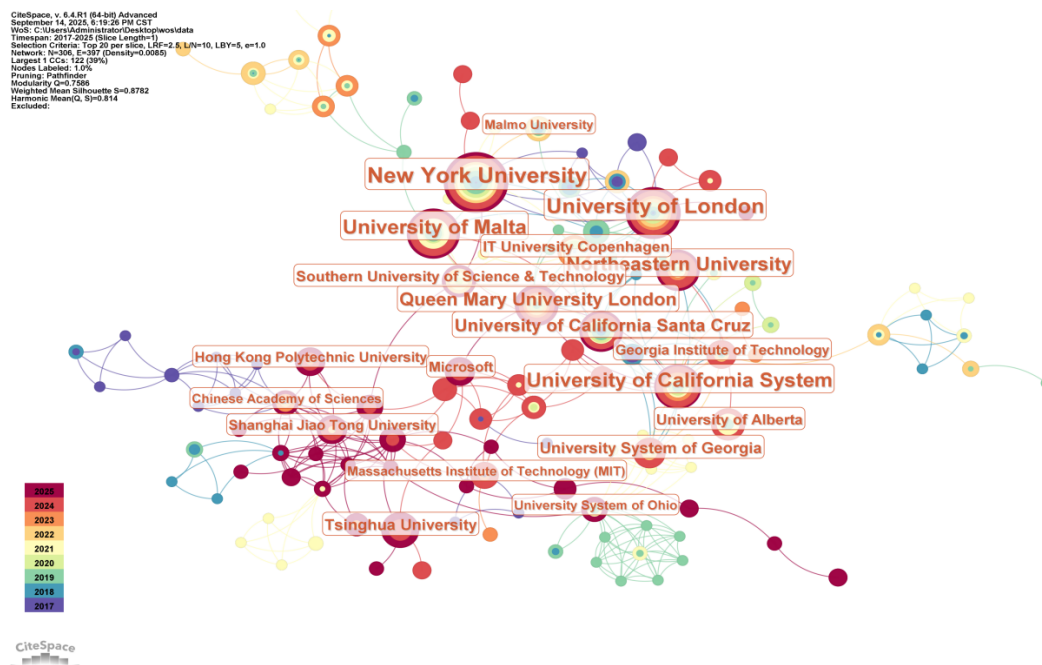


Figure 4: Institutional cooperation network

the same institution, Georgios N. Yannakakis (14 papers), and Jialin Liu (10 papers) from Lingnan University in Hong Kong.

By analyzing authors' node degree centrality (a measure of collaboration scale) and publication volume, we identify the following core scholars: Julian Togelius (New York University), who holds the central position in the network with the largest node size, focuses on procedural content generation and game AI algorithms; Georgios N. Yannakakis (University of Malta), the second-largest node, specializes in player experience modeling and affective computing. Notably, both Yannakakis and Togelius co-authored the landmark textbook *Artificial Intelligence and Games* in this field; Antonios Liapis (University of Malta) collaborates closely with Yannakakis, specializing in hybrid heuristic algorithms for game design [16].

These core authors have collaborated to produce numerous landmark research and educational achievements. A prime example is the textbook *Artificial Intelligence and Games*, co-authored by Julian Togelius and Georgios N. Yannakakis [17], which has become an authoritative reference in the field. This work not only established “game AI” as a systematic discipline with coherent theories and methodologies but also marked its evolution from fragmented research into a cohesive academic domain. Furthermore, their initiatives like the International Summer School on Artificial Intelligence and Gaming have attracted participants from dozens of countries, effectively fostering and expanding the global academic community in this field.

Taking the collaborative relationship between Julian Togelius and Georgios N. Yannakakis as an example, we conducted a focused analysis of their research partnership. Through further archival research, we discovered that as early as 2009 and 2011, the two scholars had demonstrated remarkable foresight by co-authoring

two groundbreaking papers titled *Real-Time Adaptation of Video Games via Player Experience Modeling* and *Experience-Driven Procedural Content Generation* [18]. The former proposed a closed-loop framework for real-time modeling of player emotional experiences to drive dynamic game content adjustments, laying foundational groundwork for intelligent personalized gaming experiences. The latter introduced the core paradigm of “Experience-Driven Procedural Content Generation” (EDPCG), elevating PCG research to new heights. In 2021, they published *Dungeon Crawl Stone Soup as a Domain for AI Research*, proposing a complex text-based roguelike game as an AI testing platform to encourage research on long-term planning and high-dimensional state management in artificial intelligence [19]. Their latest paper, *The Ethics of AI in Games* turns to critical thinking about technology, marking the expansion of their research vision from “what AI can do” to “how AI should do it” in terms of ethics and responsibility [20].

The collaborative framework illustrated by the aforementioned authors serves as a prime example, clearly demonstrating the evolving research trajectories across different phases—from technical refinement and application expansion to contemporary humanities and ethical considerations. This reflects how author collaborations in this field have developed an idealized model: starting with solving concrete problems, gradually establishing paradigms, tools, and communities, ultimately defining disciplinary boundaries and guiding ethical discourse. In an emerging domain, such well-established collaborative dynamics are particularly hard-won and highly valuable.

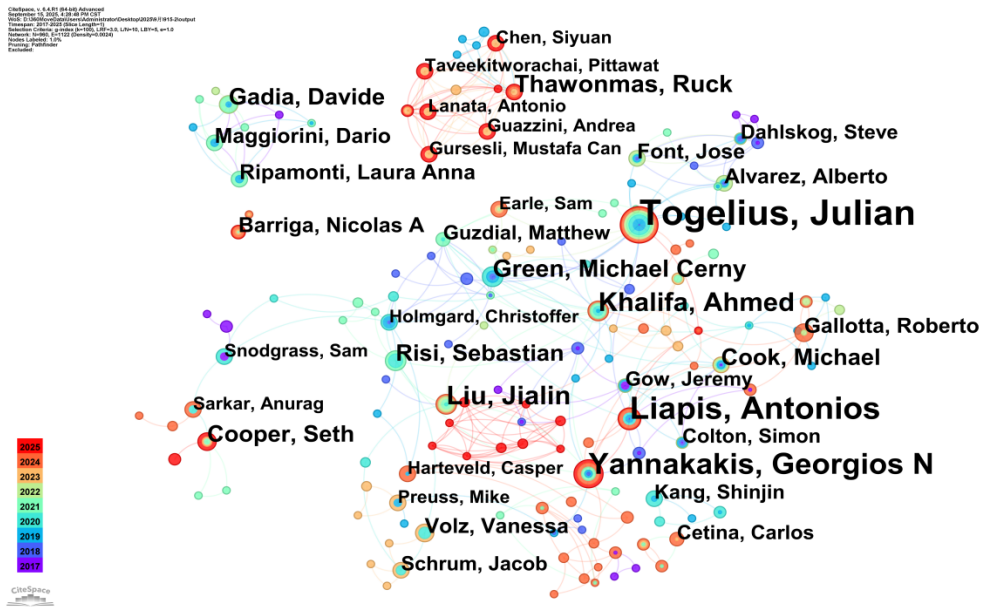


Figure 5: Author collaboration network

3.4 Keyword co-occurrence

The visualization parameters for the keyword co-occurrence network in this section are as follows: The analysis was conducted using the CiteSpace v.6.4 R1 scientometrics tool, with a time span of 2017–2025 (time slice length of 1 year) and network pruning performed using the Pathfinder algorithm. Selection criteria included the top 20 high-frequency keywords in each time slice, along with threshold parameters set at LRF=2.5, L/N=10, LBY=5, and $e=1.0$. This resulted in a maximum connected component containing 123 nodes. The clustering modularity Q-value reached 0.9255, with weighted average contour values (S) and harmonic mean (Q, S) both achieving 0.9521, indicating statistically significant and highly reliable cluster results.

In the network diagram, each node represents a keyword, with its size proportional to its citation frequency in literature [21]. The connections and color depth between nodes reflect co-occurrence intensity and temporal evolution patterns of keywords. This visualization method enables clear identification of core research themes, knowledge structures, and development trends within the field.

The co-occurrence network analysis of keywords reveals a technology-driven diversification in the research themes. As the Figure 6 shown, the core node “procedural content generation” emerges as the central hub with 0.17 centrality and 251 citations, forming the most extensive knowledge network that aligns with the substantial demand for content production in traditional game development processes. This keyword maintains close connections with both conventional “genetic algorithms” and emerging “generative adversarial networks”, underscoring its key role in technological evolution. Beyond this core node, key technical nodes like “deep learning” (48 citations), “generative adversarial networks” (46 citations), and “large language models” (44 citations) demonstrate

significant scale, reflecting a multi-technology co-development paradigm in the field.

The evolution of research hotspots over time (as evidenced by the color transition from cool to warm tones in nodes) reveals distinct technological iteration patterns. In the early phase (2017–2019, blue-toned nodes), traditional algorithms like “genetic algorithm” and “evolutionary algorithm” dominated [22]. The mid-phase (2020–2022, green-toned nodes) saw deep learning technologies such as “deep learning” and “generative adversarial networks” take center stage [23]. Recent developments (2023–2025, red-toned nodes) have witnessed emerging application-focused research themes including “dynamic difficulty adjustment” and “serious games”, demonstrating a strategic shift from content generation to experience optimization [24]. This progression reflects the gaming industry’s evolving trend toward player-centric development.

Compared to traditional single-domain technologies, generative AI game development demonstrates a more open and dynamic keyword ecosystem. The core keywords encompass both fundamental technological research and practical application scenarios, vividly reflecting the field’s interdisciplinary nature [25]. This creates a mutually reinforcing development pattern where technological innovation and real-world applications drive each other forward.

3.5 Cluster analysis of keywords

This section conducted a keyword clustering analysis of literature on generative artificial intelligence in game development, identifying distinct clusters through the Log-Likelihood (LLR) method. The cluster modularity Q-value reached 0.8782, with weighted average contour values ($S=0.9602$) and harmonic mean ($Q,S=0.814$)—all exceeding the 0.3 threshold for statistical significance. These metrics confirm the robustness of the clustering results, ensuring reliable and accurate analytical outcomes.

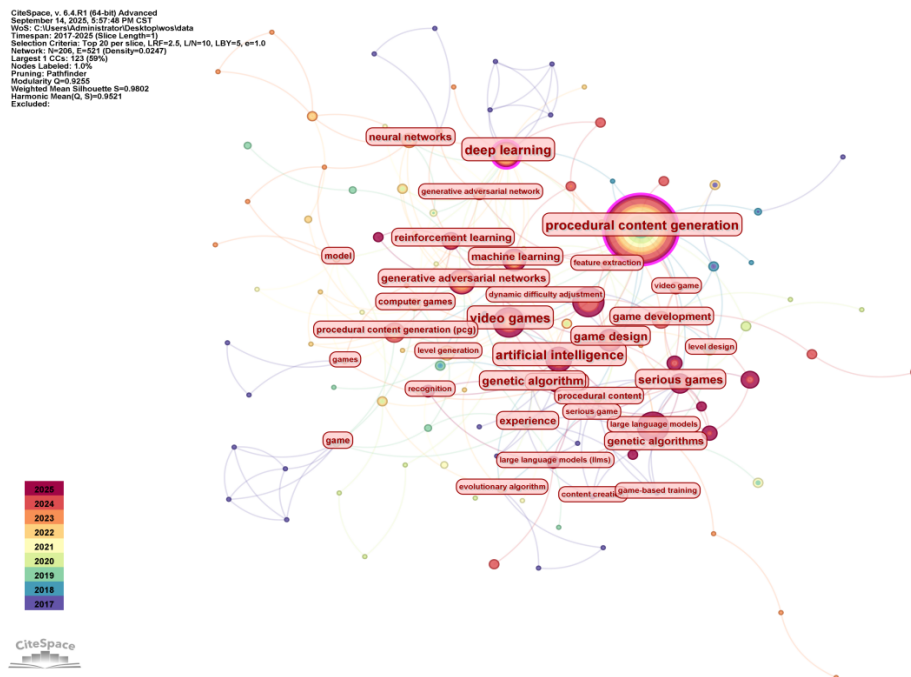


Figure 6: Keyword co-occurrence network

Through rigorous screening, six representative core research domains have been identified as shown in the Figure 7 Generative Adversarial Networks (GAN), Deep Learning, Virtual Reality, Genetic Algorithms, Level Generation, and Game Experience. The GAN domain (Cluster #0) primarily focuses on generative modeling technologies, encompassing keywords such as “procedural content generation”, “generative adversarial networks”, “image generation”, and “style transfer”. Proposed by Ian Goodfellow in 2014, GAN is an unsupervised deep learning model that demonstrates exceptional capabilities in image generation, restoration, transformation, and text generation [26]. Between 2019 and 2021, this field garnered significant attention, featuring the largest node scale and dense connectivity, indicating high technical maturity.

The deep learning field (Cluster #1) focuses on the application of neural network technologies, encompassing keywords such as “deep learning”, “neural networks”, and “convolutional neural networks” [27]. This domain maintains steady research momentum, demonstrating a progressive relationship with generative adversarial networks through “foundational frameworks-specialized applications”, collectively driving the intelligent transformation of game development [28].

The Virtual Reality (VR) field (Cluster #2) focuses on immersive gaming experiences, encompassing keywords such as “virtual reality”, “augmented reality”, and “mixed reality” [29]. VR technology is a comprehensive system that utilizes computer-generated systems to create and experience virtual worlds. By generating simulated environments through computing, it immerses users in these spaces where they can perceive multiple sensory inputs including auditory, visual, and tactile elements [30]. The warm-toned color scheme of

this cluster node reflects sustained research activity in recent years, resonating with emerging concepts like the metaverse.

The field of genetic algorithms (#3 clustering) represents the traditional algorithm path, covering keywords such as “genetic algorithm”, “evolutionary algorithm”, and “optimization” [31]. Although the node scale is relatively small, it maintains connections with emerging technologies, reflecting the continuity of the technological path.

The Level Generation Domain (#8 Cluster) focuses on game content creation, including keywords such as “level generation” and “game design” [32]. As the primary application layer for generative AI in game development, this domain establishes extensive connections with multiple technical clusters [33].

The Game Experience domain (Clusters #4 and #5) focuses on player interaction experiences, encompassing keywords such as “player experience”, “user experience”, “cognitive impact”, and “engagement” [34]. These cluster nodes are colored in a fresh hue, representing research directions within the influence layer. They prioritize the ultimate service recipient—“human beings”, concentrating on leveraging generative AI to understand and enhance player experiences, ensuring that final application outputs lean toward user experience optimization [35].

These clusters form a complete research chain spanning from foundational technologies to top-tier applications, collectively illustrating the parallel technological evolution path of traditional algorithms (genetic algorithms) and modern AI (deep learning, GAN). Different algorithms have found their roles in various stages of game development through their unique advantages, demonstrating strong problem-driven characteristics. Meanwhile, the

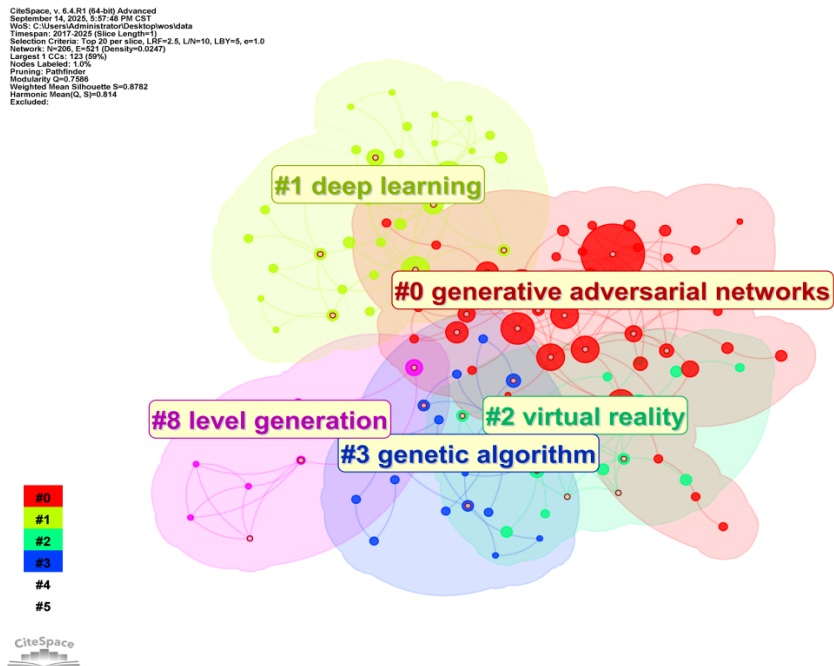


Figure 7: Keyword clustering

dense connections and intersections between these clusters reflect the research trend of transitioning from single technological breakthroughs to cross-disciplinary integration innovation. Future developments will continue along the main themes of “experience-driven” and “technology convergence,” profoundly transforming game creation methods and player experiences.

3.6 Research trend analysis

The cutting-edge research focus of generative artificial intelligence in game development has been dynamically evolving alongside breakthroughs in foundational technologies and shifting industry demands. By analyzing the temporal evolution trajectory of research frontiers from 2017 to 2025 shown in Figure 8, we can clearly observe the iterative process and evolutionary characteristics of technological pathways. Based on the prominence and temporal distribution patterns of keywords, we have divided the research cycle into three key phases (2017-2019, 2020-2022, and 2023-2025).

During Phase I (2017-2019), traditional technologies such as genetic algorithms, procedural content generation, and feature extraction dominated the field. For instance, Hede et al. (2018) systematically explored algorithmic applications in game level generation through keywords like “fitness function,” “evolution strategy,” and “optimization.” By 2019, research focus began shifting toward data-driven approaches, with keywords like “neural networks” and “deep learning” showing significant frequency increases, signaling a paradigm shift in technological development [36]. As the first stage reached maturity, the release of GPT-3 in 2020 marked a broader turning point in the development of generative AI, revealing the transformative potential of large-scale models [37]. It not only

reshaped the overall research overview but also inspired forward-looking applications in fields such as game design and interactive content generation.

During Phase Two (2020-2022), technological development accelerated significantly. Notably, the official launch of ChatGPT in 2022 marked a key milestone that accelerated the adoption of generative AI technology [38]. This period witnessed two major research trajectories: First, foundational technological breakthroughs manifested through parallel advancements in “generative adversarial networks,” “diffusion models,” and “neural radiance fields”. Second, application scenarios expanded with keywords like “game development”, “game design”, and “computer games” establishing intensive connections with AI technology, signaling the arrival of practical implementation [39].

Phase Three (2023-2025): Amid the sustained ripple effects of ChatGPT technology, research demonstrated deepening application characteristics. In the wave of large language models represented by ChatGPT, emerging hotspots like “large language models”, “dialogue systems”, and “personalized generation” emerged, particularly achieving significant progress in game narrative generation and intelligent NPC dialogue [40]. However, further analysis reveals that this phase saw relatively limited original technological breakthroughs, with most efforts focused on integrating and optimizing existing technologies—key terms like “multimodal fusion” and “real-time generation optimization” indicate this. This suggests the research faced practical challenges as theoretical innovation entered a plateau period.

This evolutionary trajectory demonstrates that generative AI in game development is undergoing a pivotal shift from “technological exploration” to “industrial application” [41]. Technologies

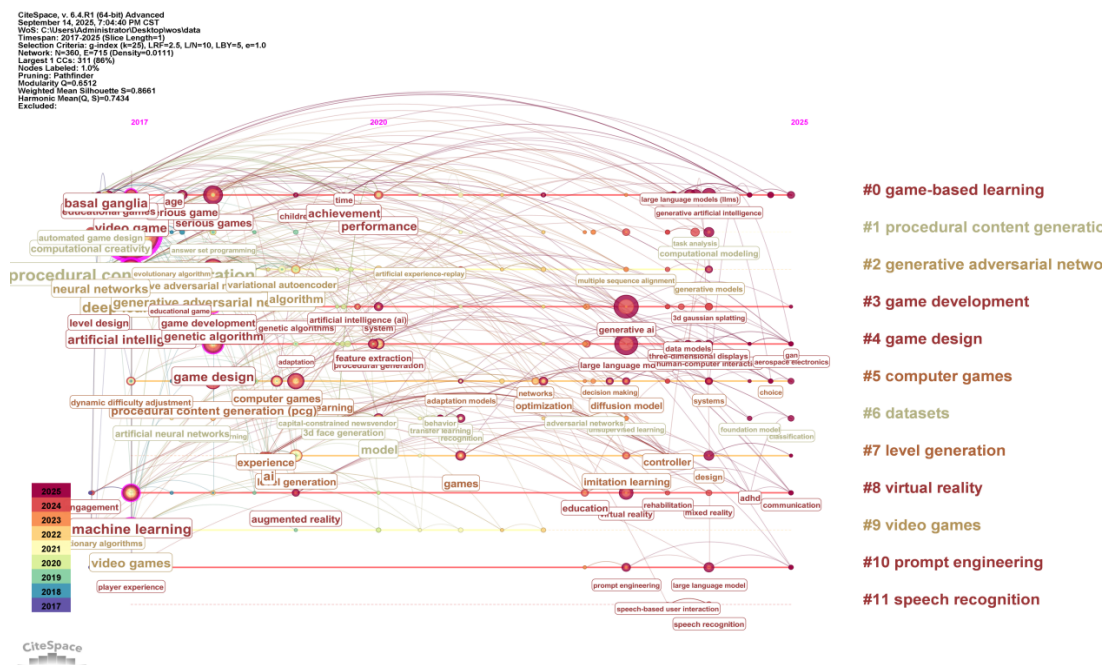


Figure 8: Keyword time zone diagram

like ChatGPT not only lower the barrier to entry but also redefine human-machine collaboration in game development [42]. Generative AI is evolving from a simple production aid into a dynamic innovation incubator, empowering creative breakthroughs with unprecedented vitality.

4 Visualization of co-cited networks

4.1 Author cited

Through author co-citation analysis, we constructed an author citation network using VOSviewer based on the frequency of co-citations among different authors in academic publications. The node sizes in the diagram typically correspond to the citation frequency of each author. Figure 9 clearly identifies several highly influential authors who serve as knowledge hubs within this field:

First are Julian Togelius and Georgios N. Yannakakis, who serve as key nodes in the author collaboration network. As the two largest and most central nodes in the co-citation network, they symbolize their core positions in the field. Their research established paradigms for Programmed Content Generation (PCG) and player experience modeling. Adam Sammerville also stands out with significant node scale, being a key highly-cited author particularly notable for his contributions to technical applications of PCG [43]. Authors like Antonios Liapis also hold substantial nodes, demonstrating their crucial contributions in specialized subfields such as game AI agents and game design tools [44].

Based on the core author nodes identified above, we can construct co-citation network clusters centered around them. The entire network is categorized into three clusters: red (theory and paradigm innovation), green (technology and tool development),

and blue (algorithmic and creative applications). The red cluster, led by Julian Togelius and Georgios N. Yannakakis, serves as the “brain” driving the field’s development. The green cluster, spearheaded by Adam Sammerville, focuses on developing tools, pipelines, and algorithms to implement theoretical frameworks. The blue cluster, anchored by Antonios Liapis, delves into specialized technical pathways, concentrating on cutting-edge applications of evolutionary algorithms and computational creativity in game design.

In conclusion, the co-citation network features indicate that generative AI game development has developed a mature discipline system with solid theory as the core, diversified technical exploration as the support, frequent internal knowledge flow, healthy structure and continuous evolution [45].

4.2 Citations of references

Figure 10 illustrates the co-citation network of publications from game development constructed using generative AI based on comprehensive literature retrieval. This network reveals knowledge connections and thematic evolution patterns among academic achievements. The network, generated through the CiteSpace clustering algorithm, exhibits significant modular characteristics with a modularity value of 0.8374 and a weighted average contour coefficient of 0.9241, demonstrating robust structural significance and stability in the clustering results. Different colored regions in the figure reflect the temporal evolution of research themes: blue nodes represent early-stage studies, green nodes correspond to mature technology phases, while yellow and red nodes mark emerging research frontiers in recent years.

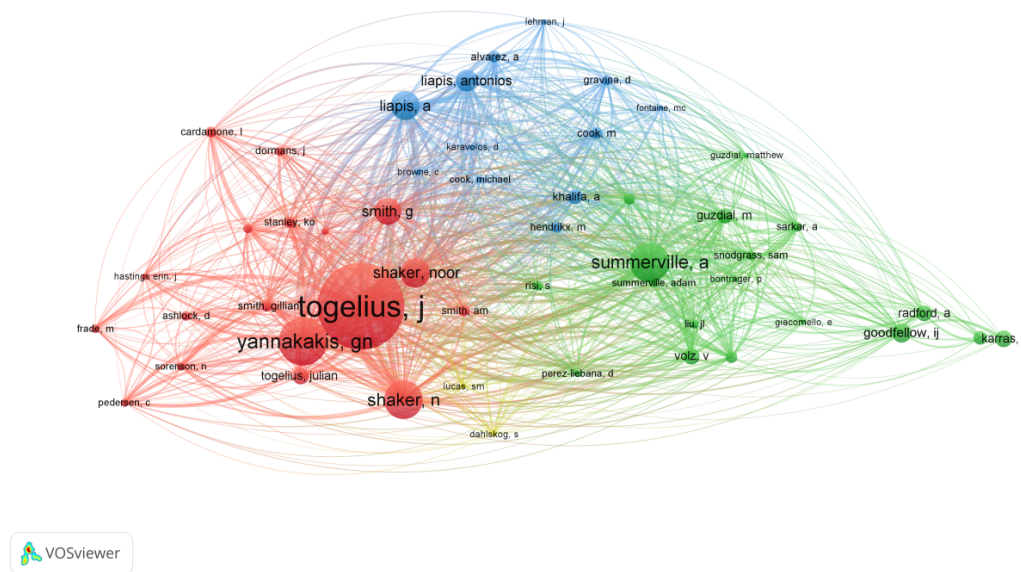


Figure 9: Author collaboration network

From the perspective of clustering structures, the network exhibits a multi-core parallel knowledge architecture. Early clusters centered on procedural content generation (PCG) (Cluster #0), which together with experience-driven content generation (Experience-Driven PCG) formed the theoretical foundation of the field. Subsequently, research focus gradually shifted to deep learning and generative adversarial networks (Cluster #1), establishing key technological pathways for generating AI game development. After 2020, the emergence of large language models and multi-modal generation (Cluster #2) marked a paradigm shift from algorithm optimization to model integration and cross-modal fusion [46]. In recent years, intelligent NPC interaction systems (Cluster #4) and player experience modeling (Cluster #5) have emerged as new growth areas, indicating that research is transitioning from technology-oriented approaches to a comprehensive development stage centered on experience and interaction [47].

Five highly cited papers were highlighted in the co-cited network, and the node centrality of these papers was all over 0.1, forming the most influential knowledge base in this field:

(i) Togelius and Yannakakis (2018) *Artificial Intelligence and Games* is a milestone work on the field of generative AI in games. As mentioned earlier, it will not be repeated here.

(ii) Summerville et al. (2018) pioneered the Programmed Content Generation through Machine Learning (PCGML) framework in their seminal work *Procedural Content Generation via Machine Learning* [48], marking a paradigm shift from heuristic algorithms to data-driven models in content generation research. By synthesizing insights into the application potential of genetic algorithms,

deep learning, and reinforcement learning across game development dimensions—including level design, narrative generation, and music composition—the authors developed a systematic solution for AI-powered intelligent content creation. This groundbreaking study not only established theoretical foundations for subsequent research but also laid the academic groundwork for integrating generative models into creative workflows, exerting significant influence on AI-driven game design methodologies.

(iii) The groundbreaking work *NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis* by Mildenhall et al. (2020) pioneered neural radiance field-based 3D reconstruction and scene generation technology [49]. By employing implicit volumetric rendering, the NeRF model achieves high-precision perspective synthesis, providing revolutionary tools for automated modeling and virtual environment generation in game development. This research propelled generative AI's technological leap from 2D images to 3D space, with its concepts being widely adopted in virtual reality (VR), digital twin technology, and interactive scene construction. It has become one of the most crucial technical foundations for generative content generation in recent years.

(iv) OpenAI's 2023 *GPT-4 Technical Report* represents a groundbreaking advancement in large language model technology for natural language generation and interactive intelligent systems [50]. By integrating multi-layer Transformer architectures with cross-modal alignment mechanisms, GPT-4 achieves unprecedented performance improvements in semantic understanding and contextual generation [51]. This report not only drives the development of AI dialogue and narrative generation technologies but also provides innovative approaches for intelligent NPCs, dynamic story

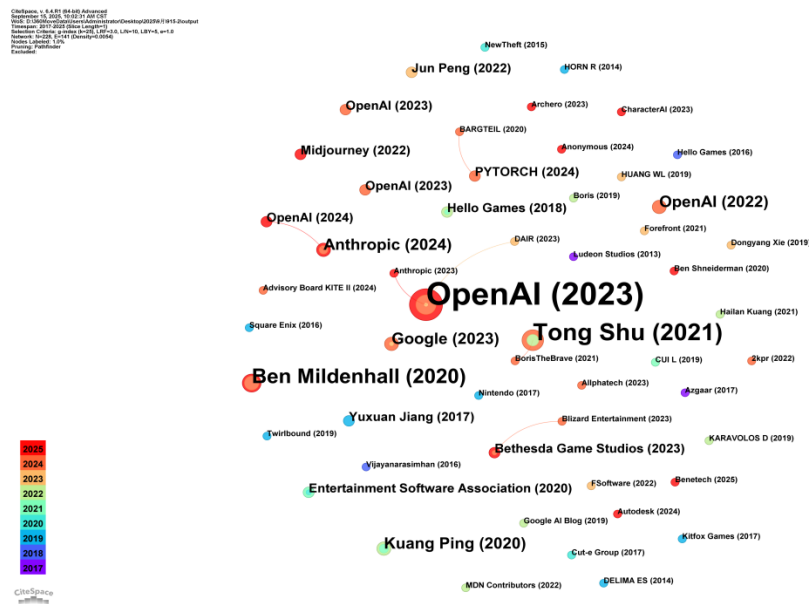


Figure 10: Literature co-citation network

generation, and personalized gaming experiences. Its release marks a new phase where generative AI transitions from laboratory research to practical applications in gaming, profoundly impacting human-computer interaction models within the industry.

(v) Anthropic’s 2024 report *Constitutional AI: Harmlessness from AI Feedback* made groundbreaking progress in alignment and ethical controllability research, providing theoretical foundations for the safe and responsible use of AI-generated content [52]. The report introduced the concept of “Constitutional AI,” which establishes rule-based self-regulation mechanisms to ensure models adhere to ethical and safety standards during content generation. This philosophy proves particularly crucial for game AI development, offering new solutions to address ethical risks, player trust issues, and content moderation challenges in narrative generation and behavioral modeling. It also marks a key shift in generative AI research toward a phase of “responsible innovation.”

These seminal studies span three key dimensions: theoretical foundations, algorithmic innovations, and application expansions, collectively establishing a comprehensive knowledge framework for generative AI-driven game development. The research trajectory has fundamentally shifted from “algorithm-centric” approaches to “experience-centric” strategies, where technological breakthroughs, creative designs, and human-computer interactions form an interdependent dynamic system. This evolution lays a robust academic foundation for future intelligent and personalized game development.

5 Discussion and conclusions

5.1 Key findings

Research on generative AI in game development has risen steadily since 2017, accelerating after 2020 as large-scale generative models began reshaping creative industries [53]. The focus has shifted from

procedural content generation (PCG) to data-driven and experience-oriented approaches, showing that generative AI has become central to intelligent and creative game development.

Keyword analysis identifies PCG, machine learning, reinforcement learning, game design, and player experience as major themes, with recent growth in deep reinforcement learning, neural networks, and GANs. These changes mark a move toward data-driven creativity and stronger links among AI, interaction, and narrative design.

Three phases can be observed. From 2017 to 2019, studies emphasized rule-based PCG and reinforcement learning. Between 2020 and 2022, the release of GPT-3 signaled a shift from general modeling to domain-specific applications such as game design. Since 2023, research has focused on multimodal generation and human-AI co-creation, reflecting the turn from algorithmic control to collaborative creativity [54].

Influential scholars include Julian Togelius, Georgios N. Yannakakis, and Noor Shaker, who built the foundations of PCG and computational creativity, while Antonios Liapis and Mark J. Nelson extended these ideas to adaptive design and narrative generation. In China, Li Deyi, Song Jianwu, and Chen Xiangming advanced intelligent game systems and creative computing, promoting local development and interdisciplinary integration.

At the national and institutional levels, the United States and China lead in publication output, followed by the United Kingdom, Germany, and Canada. Collaboration networks form a multi-centered structure dominated by North America and East Asia, with growing participation from Europe. Representative institutions include Carnegie Mellon University, Tsinghua University, and the University of Southampton, which serve as core nodes connecting academic and industrial research. Keyword evolution—from

“generation” and “automation” to “creativity,” “interaction,” and “experience”—shows the rise of user-centered and co-creative design paradigms.

Overall, research has expanded from technical journals to interdisciplinary venues combining technology, art, and media, evolving from a niche topic into a mature, globally relevant field.

5.2 Practical significance

Research in this field provides technical support for optimizing and innovating the gaming industry. Generative AI technology is reshaping game development workflows: programmatic content generation can reduce production costs by up to 70%, dynamic difficulty adjustment techniques boost player retention rates by 25–40%, while automated testing reduces manual testing expenses by over 60%. These innovations enable small and medium-sized studios to develop high-quality games at lower costs, easing developers’ workload while fostering a more diversified innovation ecosystem in the industry [55].

Meanwhile, the optimization and restructuring of game development through generative AI can in turn enrich AI research. The gaming environment provides an ideal experimental platform for AI studies [56]. The complexity of game worlds, their clearly defined rules, and controllable states make them a “digital wind tunnel” for training and testing AI algorithms. The methodologies developed in game-based generative AI can be transferred to industrial applications such as digital twins and virtual simulations, driving AI technology’s expansion from entertainment to industrial applications [57].

Finally, the effective application of generative AI in game development requires strong societal support [58]. In talent cultivation, we should nurture interdisciplinary professionals with both technical expertise and artistic literacy [59]. Regarding standardization, industry organizations should be encouraged to establish quality standards for generated content, ethical evaluation frameworks, and technical application guidelines to steer responsible technological development. For industrial support, it is recommended to create dedicated funds to demonstrate the application of generative AI technology in areas such as indie game development and digital transformation of traditional culture.

5.3 Limitations and prospects

This study utilizes Web of Science English literature data, which inherently limits data coverage. While encompassing mainstream international research findings, it may exclude multilingual publications (such as Chinese and Japanese game development studies) and unpublished industry technical reports. Future research could integrate multilingual datasets with industry white papers to establish a more comprehensive analytical foundation.

Meanwhile, this study demonstrates room for methodological refinement. While current quantitative approaches can reveal macro-level trends, they lack depth in interpreting the underlying semantics of research content. Future investigations should integrate natural language processing (NLP) technologies to conduct thematic modeling and semantic analysis of literature, enabling more precise tracking of technological evolution trajectories. Additionally,

strengthening case studies would facilitate comprehensive analysis of specific technologies’ transition from laboratory research to industrial implementation.

Finally, this study primarily adopts a technical perspective while lacking integration of social science approaches. As an emerging field, the integration of generative AI technology with traditional game development industries will inevitably bring about various challenges and issues, including copyright ownership and profit distribution mechanisms for AI-generated content, as well as the impact of generative AI on game developers’ professional ecosystems. Addressing these issues will significantly influence the healthy development of technology and the maintenance of a healthy industry environment.

6 Conclusion

Based on the systematic review and analysis of the relevant literature on generative artificial intelligence in the field of game development, this study clearly reveals the internal mechanism and development trajectory of the evolution of this technology from an auxiliary tool to a creative collaborative subject.

From a developmental perspective, the deep integration of generative AI with game development carries profound implications that transcend individual industry transformations. Generative AI is reshaping the production workflows of the gaming sector, evolving from basic production tools to intelligent innovation infrastructure that enables large-scale personalized game creation. Meanwhile, gaming environments—acting as highly complex “digital wind tunnels”—provide an ideal testing ground for training and validating generative AI. Breakthroughs in this field will spread through knowledge transfer mechanisms from the entertainment industry to broader application scenarios like autonomous driving and digital twins.

Ultimately, the fundamental value of this technological evolution lies in building a co-evolutionary paradigm of human creativity and machine intelligence, jointly expanding the boundaries of cognition, and pushing the interactive entertainment industry into a new stage of development.

References

- [1] S. Risi and J. Togelius, “Increasing generality in machine learning through procedural content generation,” *Nature Machine Intelligence*, vol. 2, no. 8, pp. 428–436, Aug. 2020, doi: 10.1038/s42256-020-0208-z.
- [2] J. Liu, S. Snodgrass, A. Khalifa, S. Risi, G. N. Yannakakis, and J. Togelius, “Deep learning for procedural content generation,” *Neural Computing and Applications*, vol. 33, no. 1, pp. 19–37, Oct. 2020, doi: 10.1007/s00521-020-05383-8.
- [3] B. Zada, M. Kwon, and S.-W. Kim, “Current Trends in Acetins Production: Green versus Non-Green Synthesis,” *Molecules*, vol. 27, no. 7, p. 2255, Mar. 2022, doi: 10.3390/molecules27072255.
- [4] A. Summerville *et al.*, “Procedural Content Generation via Machine Learning (PCGML),” *IEEE Transactions on Games*, vol. 10, no. 3, pp. 257–270, Jun. 2018, doi: 10.1109/tg.2018.2846639.
- [5] K. Fukaya, D. Daylamani-Zad, and H. W. Agius, “Intelligent graphical game asset generation: A conceptual framework and systematic review of the state of the art,” *arXiv preprint*, 2023. [Online]. Available: <https://arxiv.org/abs/2311.10129>.
- [6] C. Hu, Y. Zhao and J. Liu, “Game Generation via Large Language Models,” *2024 IEEE Conference on Games (CoG)*, Milan, Italy, 2024, pp. 1–4, doi: 10.1109/CoG60054.2024.10645597.
- [7] T. Sakirin and S. Kusuma, “A survey of Generative Artificial intelligence techniques,” *Deleted Journal*, vol. 2023, pp. 10–14, Mar. 2023, doi: 10.58496/bjai/2023/003.
- [8] C. V. Oliveira and P. N. Rito, “Enhancing player experience through generative Artificial intelligence: custom interaction in game design,” in *Communications*

- in computer and information science, 2025, pp. 259–268. doi: 10.1007/978-3-031-81713-7_18.
- [9] Y. Zakaria, M. Fayek, and M. Hadhoud, “Procedural level Generation for Sokoban via Deep Learning: an Experimental study,” *IEEE Transactions on Games*, vol. 15, no. 1, pp. 108–120, May 2022, doi: 10.1109/tg.2022.3175795.
 - [10] R. Gallotta, A. Liapis, and G. Yannakakis, “Consistent Game Content Creation via Function Calling for Large Language Models,” *2024 IEEE Conference on Games (CoG)*, Milan, Italy, 2024, pp. 1–4, doi: 10.1109/CoG60054.2024.10645599.
 - [11] G. Todd and Others, “Level Generation Through Large Language Models,” in *ACM Proceedings*, 2023.
 - [12] V. Geroimenko, “Generative AI: From Human–Computer interaction to Human–Computer Creativity,” in *Springer series on cultural computing*, 2025, pp. 3–29, doi: 10.1007/978-3-031-86551-0_1.
 - [13] M. B. Tavares, *Aprendizagem por Reforço Aplicada a Jogos de Luta*, Universidade de Aveiro, Portugal, 2025.
 - [14] M. Sedighi, “Application of word co-occurrence analysis method in mapping of the scientific fields (case study: the field of Informetrics),” *Library Review*, vol. 65, no. 1/2, pp. 52–64, Jan. 2016, doi: 10.1108/Lr-07-2015-0075.
 - [15] M. F. Maleki and R. Zhao, “Procedural Content Generation in Games: A Survey with Insights on Emerging LLM Integration,” *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, vol. 20, no. 1, pp. 167–178, Nov. 2024, doi: 10.1609/aiide.v20i1.31877.
 - [16] M. Barthet, M. Kaselimi, K. Pinitas, K. Makantasis, A. Liapis, and G. N. Yannakakis, “GameVibe: a multimodal affective game corpus,” *Scientific Data*, vol. 11, no. 1, Nov. 2024, doi: 10.1038/s41597-024-04022-4.
 - [17] G. N. Yannakakis and J. Togelius, *Artificial Intelligence and Games*, Springer Science and Business Media LLC, 2018.
 - [18] G. N. Yannakakis and J. Togelius, “Experience-Driven procedural content generation,” *IEEE Transactions on Affective Computing*, vol. 2, no. 3, pp. 147–161, Apr. 2011, doi: 10.1109/t-affc.2011.6.
 - [19] D. Dannenhauer, M. W. Floyd, J. Decker, and D. W. Aha, “Dungeon Crawl Stone Soup as an evaluation domain for artificial intelligence,” *arXiv (Cornell University)*, Jan. 2019, doi: 10.48550/arxiv.1902.01769.
 - [20] D. Melhart, J. Togelius, B. Mikkelsen, C. Holmgård, and G. N. Yannakakis, “The ethics of AI in games,” *IEEE Transactions on Affective Computing*, vol. 15, no. 1, pp. 79–92, May 2023, doi: 10.1109/taffc.2023.3276425.
 - [21] M. M. Mostafa, “Twenty years of Wikipedia in scholarly publications: a bibliometric network analysis of the thematic and citation landscape,” *Quality & Quantity*, vol. 57, no. 6, pp. 5623–5653, Feb. 2023, doi: 10.1007/s11355-023-01626-7.
 - [22] J. De Miguel and J. Á. Velázquez-Iturbide, “What is an algorithm?,” in *Advances in information quality and management*, 2024, pp. 1–17, doi: 10.4018/978-1-6684-7366-5.ch072.
 - [23] I. Goodfellow et al., “Generative adversarial networks,” *Communications of the ACM*, vol. 63, no. 11, pp. 139–144, Oct. 2020, doi: 10.1145/3422622.
 - [24] P. D. Paraschos and D. E. Koulouriotis, “Game Difficulty adaptation and Experience Personalization: A Literature review,” *International Journal of Human-Computer Interaction*, vol. 39, no. 1, pp. 1–22, Apr. 2022, doi: 10.1080/10447318.2021.2020008.
 - [25] S. Werning, “Generative AI and the technological imaginary of game design,” in *Creative working lives*, 2024, pp. 67–90, doi: 10.1007/978-3-031-45693-0_4.
 - [26] R. A. R. Sriram, L. K. M., and S. Sudarsan, “A review of GAN-based methods for image translation and caption generation,” *2022 6th International Conference on Computing Methodologies and Communication (ICCMC)*, pp. 1160–1170, Jul. 2025, doi: 10.1109/iccmc65190.2025.11140882.
 - [27] A. K. S. Alsajri and A. V. Hacimahmud, “Review of deep learning: Convolutional Neural Network Algorithm,” *Deleted Journal*, vol. 2023, pp. 19–25, Apr. 2023, doi: 10.58496/bjml/2023/004.
 - [28] A. Aggarwal, M. Mittal, and G. Battineni, “Generative adversarial network: An overview of theory and applications,” *International Journal of Information Management Data Insights*, vol. 1, no. 1, p. 100004, Jan. 2021, doi: 10.1016/j.jjimei.2020.100004.
 - [29] M. Lo, B. C. Wünsche, M. Manfredini, and S. Hooper, “(Semi-)immersive digital placemaking: A systematic literature review on augmented reality, virtual reality, and mixed reality applications in placemaking,” *Convergence the International Journal of Research Into New Media Technologies*, Aug. 2025, doi: 10.1177/13548565251372796.
 - [30] J. Rubio-Tamayo, M. G. Barrio, and F. G. García, “Immersive Environments and Virtual Reality: Systematic review and advances in communication, interaction and simulation,” *Multimodal Technologies and Interaction*, vol. 1, no. 4, p. 21, Sep. 2017, doi: 10.3390/mti1040021.
 - [31] A. J. Fofanah, S. Koroma, and H. I. Bangura, “Experimental Exploration of Evolutionary Algorithms and their Applications in Complex Problems: Genetic Algorithm and Particle Swarm Optimization Algorithm,” *British Journal of Healthcare and Medical Research*, vol. 10, no. 2, Apr. 2023, doi: 10.14738/bjhm.102.14427.
 - [32] S. Alyaseri, “Generative AI in Procedural Content Generation for Computer Games: Key contributions and Trends,” *2017 IEEE Region 10 Symposium (TEN-SYMP)*, pp. 1–8, Jul. 2025, doi: 10.1109/tensymp63728.2025.11144949.
 - [33] Z. Lv, “Generative artificial intelligence in the metaverse era,” *Cognitive Robotics*, vol. 3, pp. 208–217, Jan. 2023, doi: 10.1016/j.cogr.2023.06.001.
 - [34] Z. Diao, P. Meng, X. Meng, and L. Zhang, “The transfer of In-Game behaviors and emotions to Real-World experiences in Game World,” *Behavioral Sciences*, vol. 15, no. 9, p. 1203, Sep. 2025, doi: 10.3390/bs15091203.
 - [35] C. V. Oliveira and P. N. Rito, “Enhancing player experience through generative Artificial intelligence: custom interaction in game design,” in *Communications in computer and information science*, 2025, pp. 259–268, doi: 10.1007/978-3-031-81713-7_18.
 - [36] S. Dargan, M. Kumar, M. R. Ayyagari, and G. Kumar, “A survey of Deep Learning and its Applications: A New Paradigm to Machine Learning,” *Archives of Computational Methods in Engineering*, vol. 27, no. 4, pp. 1071–1092, Jun. 2019, doi: 10.1007/s11831-019-09344-w.
 - [37] N. El-Bassiouny, W. Amann, D. El-Bassiouny, and C. Hauser, *Artificial Intelligence and Responsible Management Education – Current Applications and Future Directions*, Routledge, 2025.
 - [38] A. S. Alalqa, “The history of the artificial intelligence revolution and the nature of generative AI work,” *Deleted Journal*, vol. 2, no. 4, pp. 1–24, Jan. 2024, doi: 10.59232/air-v2i3p101.
 - [39] A. Geriş et al., “Revolutionising game design,” *DergiPark (Istanbul University)*, Jun. 2024, doi: 10.14783/maruoneri.1499524.
 - [40] J. Huang and K. Huang, “ChatGPT in gaming industry,” in *Future of business and finance*, 2023, pp. 243–269, doi: 10.1007/978-3-031-45282-6_9.
 - [41] Y. Song, K. Wu, and J. Ding, “Developing an immersive game-based learning platform with generative artificial intelligence and virtual reality technologies – ‘LearningverseVR,’” *Computers & Education X Reality*, vol. 4, p. 100069, Jan. 2024, doi: 10.1016/j.cexr.2024.100069.
 - [42] F. F.-H. Nah, R. Zheng, J. Cai, K. Siau, and L. Chen, “Generative AI and ChatGPT: Applications, challenges, and AI-human collaboration,” *Journal of Information Technology Case and Application Research*, vol. 25, no. 3, pp. 277–304, Jul. 2023, doi: 10.1080/15228053.2023.2233814.
 - [43] A. Summerville et al., “Procedural Content Generation via Machine Learning (PCGML),” *IEEE Transactions on Games*, vol. 10, no. 3, pp. 257–270, Jun. 2018, doi: 10.1109/tg.2018.2846639.
 - [44] A. Liapis, G. Smith, and N. Shaker, “Mixed-initiative content creation,” in *Computational synthesis and creative systems*, 2016, pp. 195–214, doi: 10.1007/978-3-319-42716-4_11.
 - [45] A. Pescapé, “Exploring the current state and future potential of generative artificial intelligence using a generative artificial intelligence,” in *Integrated science*, 2024, pp. 37–56, doi: 10.1007/978-3-031-58363-6_4.
 - [46] Z. Chen et al., “Evolution and prospects of foundation models: From large language models to large multimodal models,” *Computers, Materials & Continua (Print)*, vol. 80, no. 2, pp. 1753–1808, Jan. 2024, doi: 10.32604/cmc.2024.052618.
 - [47] X. Zhu, “Behavior tree design of intelligent behavior of non-player character (NPC) based on Unity3D,” *Journal of Intelligent & Fuzzy Systems*, vol. 37, no. 5, pp. 6071–6079, Jun. 2019, doi: 10.3233/jifs-179190.
 - [48] G. Todd, S. Earle, M. U. Nasir, M. C. Green, and J. Togelius, “Level generation through large language models,” in *Proc. 18th Int. Conf. Foundations of Digital Games (FDG)*, 2023.
 - [49] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, “NERF: Representing scenes as neural radiance fields for view synthesis,” in *Lecture notes in computer science*, 2020, pp. 405–421, doi: 10.1007/978-3-030-58452-8_24.
 - [50] Z. Wang, Z. Chu, T. V. Doan, S. Ni, M. Yang, and W. Zhang, “History, development, and principles of large language models: an introductory survey,” *AI And Ethics*, Oct. 2024, doi: 10.1007/s43681-024-00583-7.
 - [51] S. Long et al., “A Survey on Intelligent Network Operations and Performance Optimization Based on Large Language Models,” *IEEE Communications Surveys & Tutorials*, p. 1, Jan. 2025, doi: 10.1109/comst.2025.3526606.
 - [52] B. Yuntao et al., “Constitutional AI: Harmlessness from AI feedback,” *Cambridge University Press eBooks*, Dec. 15, 2022, https://arxiv.org/abs/2212.08073 (accessed Oct. 17, 2025).
 - [53] R. Gupta, S. Tiwari, and P. Chaudhary, “Generative AI techniques and models,” in *Lecture notes on data engineering and communications technologies*, 2025, pp. 45–64, doi: 10.1007/978-3-031-82062-5_3.
 - [54] V. Geroimenko, “Generative AI: From Human–Computer interaction to Human–Computer Creativity,” in *Springer series on cultural computing*, 2025, pp. 3–29, doi: 10.1007/978-3-031-86551-0_1.
 - [55] S. Werning, “Generative AI and the technological imaginary of game design,” in *Creative working lives*, 2024, pp. 67–90, doi: 10.1007/978-3-031-45693-0_4.
 - [56] W. Westera et al., “Artificial intelligence moving serious gaming: Presenting reusable game AI components,” *Education and Information Technologies*, vol. 25, no. 1, pp. 351–380, Jul. 2019, doi: 10.1007/s10639-019-09968-2.
 - [57] F. Rahimi, A. Sadeghi-Niaraki, and S.-M. Choi, “Generative AI Meets Virtual Reality: a comprehensive survey on applications, challenges, and future direction,” *IEEE Access*, p. 1, Jan. 2025, doi: 10.1109/access.2025.3574779.

- [58] W. Shafik, “Generative AI for social good and sustainable development,” in *Studies in computational intelligence*, 2024, pp. 185–217. doi: 10.1007/978-981-97-8460-8_10.
- [59] L. Qian, W. Cao, and L. Chen, “Influence of artificial intelligence on higher education reform and talent cultivation in the digital intelligence era,” *Scientific Reports*, vol. 15, no. 1, Feb. 2025, doi: 10.1038/s41598-025-89392-4.